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27 March 1969
SUPERSEDING
FA-PD-MI-2663
24 May 1965

MILITARY SPECIFICATION
THRUSTER, CARTRIDGE ACTUATED: M20A1, REMANUFACTURED METAL PARTS ASSEMBLY

1. SCOPE

1.1 This Specification covers Thruster, Cartridge Actuated: M20A1, Remanufactured Metal Parts Assembly. The assembly will ultimately be loaded and used in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for

STANDARDS
Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-1167	- Ammunition Data Cart

DRAWINGS
Munitions Command

F10533858	- Thruster, Cartridge Actuated, M20A1, Metal Parts Assembly
C8797741	- Container, Ammunition, Fiber M347 for Thruster, Cartridge Actuated, M20A1, M25 or M5A2
C8797742	- Box, Packing, Ammunition for Thruster, Cartridge Actuated M20A1, M25 or M5A2
IEL-10533858	- Inspection Equipment List for Thruster, Cartridge Actuated M20A1, Metal Parts Assembly

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(Copies of specification, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standard Association, 10 East 40th Street, New York, N.Y. 10016.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Material.- Component parts for remanufacture may be obtained from M20A1 or M20 Thruster models.

3.3 Hydrostatic pressure.- The body and head shall withstand a constant internal pressure of 8000 pounds per square inch(psi) minimum, for 15 seconds, minimum, without application of any external load and shall show no evidence of leakage or mechanical failure. (see 6.2)

3.4 Structural strength.- The assembled thruster shall withstand a static tensile force of 2000 pounds, minimum, for 15 seconds, minimum, between mounting points without unlocking and without mechanical failure. (see 6.2)

3.5 Metal parts assembly.- Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing, F10533858. All threaded joints shall be assembled with sufficient torque to lightly bottom components.

3.5.1 The remanufactured thruster metal parts assembly shall contain the following new component parts: shear pin, piston locking spring and "O" rings.

3.5.2 The remanufactured thruster metal parts assembly shall include parts that comply with the following. Any parts that do not comply with requirements shall be either modified or discarded.

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3.5.2.1 End sleeve, 8596446 - Shall have two .040 diameter holes as shown in section B-B of drawing D8596446.

3.5.2.2 Head, 10521455 - Shall be compatible for nylon thread fastening as shown in details "A" and "C" of drawing D10521455.

3.5.2.3 Piston, 10521456 - Shall be compatible for nylon thread fastening, with no set screw holes, as shown on drawing F10521456.

3.5.3 Trunnion rotation.- Upon completion of assembly, the trunnion shall be free to rotate 360 degrees.

3.5.4 End sleeve rotation.- Upon completion of assembly and after structural strength test, the end sleeve shall be free to rotate 360 degrees in a clockwise direction.

3.6 Protective finish.- Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of contour configuration. Remanufactured exterior components shall have the existing protective finish stripped and the specified protective finish reapplied. (see 6.4)

3.7 Residual magnetism.- The metal parts assembly shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches. (see 6.3)

3.8 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects.- All components shall be free from cracks, splits, or other metal defects.

3.8.2 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter.- No part or assembly shall contain chips, dirt, grease, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define Quality Assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550 except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.- Unless specified otherwise by the contracting officer, formation of lots and presentation of lots shall be in accordance with the "Submission of Product" provision of MIL-STD-105.

4.2.1.2 Size of lot.- The shipping lot of metal parts assemblies shall consist of 1200 Thrusters, maximum, provided the homogeneity of remanufacturing process has been retained throughout the inspection lot. A smaller lot may be shipped upon approval of the contracting officer.

4.2.2 Examination.- The roughness comparison specimens prescribed in ASA-B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

4.2.2.1 Reclaimed parts.- Examination of reuseable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be perfromed for all such defects. All nonconforming parts shall be reprocessed or rejected.

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4.2.2.2 Reprocessed parts.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table I and Table II. For an individual defect (single characteristic) the criteria in Table I shall be used. For total defects within a major or minor group, the criteria in Table II shall be used. All nonconforming material shall be rejected.

TABLE I

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination				
301 - 1,300	110	1	2	2	3
1,301 - 3,200	150	2	3	3	4
3,201 - 8,000	225	3	4	4	5
8,001 - 22,000	300	4	5	5	6

TABLE II

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects.....	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over 43	6	8	11	14
Sample Sizes for Minor Defects.....	**110**	**150**	**225**	**300**
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over 48	8	11	17	20

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4.2.2.3 Key, Drawing C8593857 covering a detail of Drawing F10533858.

Defects	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr or sharp corner	Visual
104. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.4 Pin, Firing, Drawing D8593864 covering a detail of Drawing F10533858.

CRITICAL	
None defined.	
MAJOR	
101. Metal defective	Visual
102. Protective finish inadequate or defective	Visual
103. Burr or sharp corner	Visual
104. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.5 Body and Pellet Assembly, Drawing D8594481 covering a detail of Drawing F10533858.

CRITICAL	
None defined.	

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4.2.2.5 (Continued)

Defects	Method of Inspection
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
106. Pellets loose, missing or improperly inserted	Visual
MINOR	
None defined.	

4.2.2.6 Trunnion, Drawing D8594482 covering a detail of Drawing F10533858.

CRITICAL	
None defined.	
MAJOR	
101. Diameter of lugs.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
MINOR	
201. Burr or sharp corner	Visual
202. Foreign matter	Visual

4.2.2.7 Buffer screw and pellet assembly, Drawing C8594486 covering a detail of Drawing F10533858.

CRITICAL	
None defined.	

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4.2.2.7 (Continued)

Defects	Method of Inspection
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
106. Pellet loose, missing or improperly inserted	Visual
MINOR	
None defined.	

4.2.2.8 Sleeve, End, Drawing, D8596446 covering a detail of Drawing F10533858.

CRITICAL	
None defined.	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.9 Head, Drawing D10521455 covering a detail of Drawing F10533858.

CRITICAL	
None defined.	

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4.2.2.9 (Continued)

Defects	Method of Inspection
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.10 Piston, Drawing F10521456 covering a detail of Drawing F10533858.

CRITICAL	
None defined.	
MAJOR	
101. Thread damaged or not full	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr or sharp corner	Visual
105. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.11 Retainer, Guide, Drawing D10521616 covering a detail of Drawing F10533858

CRITICAL	
None defined.	

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4.4.4.11 (Continued)

Defects		Method of Inspection
MAJOR		
101.	Thread damaged or not full	Visual
102.	Metal defective	Visual
103.	Protective finish inadequate or defective	Visual
104.	Burr or sharp corner	Visual
105.	Foreign matter	Visual
106.	Pellets loose, missing or improperly inserted	Visual
MINOR		
	None defined.	

4.2.2.12 Sleeve, Drawing C10521617 covering a detail of Drawing F10533858.

CRITICAL		
	None defined.	
MAJOR		
101.	Thread damaged or not full	Visual
102.	Metal defective	Visual
103.	Protective finish inadequate or defective	Visual
104.	Burr	Visual
105.	Foreign matter	Visual
MINOR		
	None defined.	

4.2.2.13 Assembly, Drawing F10533858, less all component parts except Guide Retainer 10521616, Firing Pin 8593864, O-Ring MS 29513-12.

CRITICAL		
1.	Protrusion of the firing pin	Gage

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4.2.2.13 (Continued)

Defect	Method of Inspection
MAJOR	
101. Any part missing	Visual
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
None defined.	

4.2.2.14 Metal Parts Assembly, Drawing F10533858.

CRITICAL	
None defined.	
MAJOR	
101. Trunnion not free to rotate 360 degrees	Manual
102. End sleeve not free to rotate 360 degrees clockwise	Manual
103. Any part missing, except as otherwise classified	Visual
104. Improper assembly	Visual
105. Thread damaged or not full	Visual
106. Burr	Visual
107. Foreign matter	Visual
MINOR	
201. Shipping cap or plug missing	Visual
202. Protective finish inadequate or defective	Visual

4.2.2.15 Container, Fiber Drawing C8797741.

CRITICAL	
None defined.	
MAJOR	
None defined.	

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4.2.2.15 (Continued)

Defect	Method of Inspection
MINOR	
201. Gap between body and cover, max.	Manual
202. Contents loose	Manual
203. Metal ends loose or distorted	Manual
204. Cut or damaged through all impregnated layers	Visual

4.2.2.16 Box, Packing, Drawing C8797742.

CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Manual & Visual
203. Strapping missing, loose, broken or improperly assembled	Manual & Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking misleading or unidentifiable	Visual
207. Number of units incorrect	Visual

4.2.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.3.

4.2.3.1 Hydrostatic pressure.- Each body and each head shall be subjected to an internal hydrostatic pressure. Failure of any unit to comply with the requirements of 3.3 shall be cause for rejection of the unit.

4.2.3.2 Structural strength.- Each thruster shall be subjected to the structural strength test. Failure of any thruster to comply with the requirements of 3.4 shall be cause for rejection of the thruster.

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4.2.3.3 Protective finish.- The number of units for protective finish tests shall be in accordance with the applicable MIL-Standard. Failure of any sample to comply with the requirements, shall be cause for rejection of the lot.

4.2.3.4 Residual magnetism.- Each assembly shall be subjected to the test for excessive residual magnetism. Failure of any unit to comply with the requirements of 3.7 shall be cause for rejection of the assembly.

4.2.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-10533858.

4.3 Test methods and procedures.

4.3.1 Hydrostatic pressure.- Prior to application of protective finish, the body and head, individually, shall be mounted in an approved device. The testing device used shall internally seal the body and head, as applicable, at each end without applying any axial load other than that applied by the liquid pressure.

4.3.2 Structural strength.- The thruster shall be subjected to the structural strength test in accordance with the requirements of 3.4. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.3.3 Protective finish.- Test procedures for protective finish shall be in accordance with the specification referenced on the component drawing.

4.3.4 Residual magnetism.- The thruster shall be tested for excessive residual magnetism in an area free of local magnetic effects by placing the thruster 6 inches from and at the same height as an approved compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The thruster shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. The procedure shall be repeated for each 90 degree rotation of the thruster.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Level A.- Packing and marking shall be in accordance with Drawings C8797741 and C8797742.

5.2 Data cards.- Data card information shall be as specified in MIL-STD-1167.

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6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 When remanufacturing and loading of assembly is performed by the same facility, the requirement of 3.7 is waived and the residual magnetism performed on the loaded assembly only.

6.4 Special instructions covering the modification, repair and assembly of remanufactured components shall be as follows:

6.4.1 Protective finish.- Remanufactured components which include Head, Body, Trunnion and End sleeve shall be stripped of protective finish and re-plated in accordance with the applicable drawings and specification.

6.4.2 Nylon pellets.- Parts which contain nylon pellets and are to be replated, shall have the pellets removed and new pellets installed after re-plating in accordance with the applicable drawings.

6.5 Supersession data.- This specification including the requirements of Purchase Description FA-PD-MI-2663 dated 24 May 1965.

CUSTODIAN:

Army - MU

PREPARING ACTIVITY:

Army - MU

Project No.1377-A193

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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